Linux Administration Lab Guide

Module 13: Introduction to Firewalld — Step-by-Step Task Solutions

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Task 1: Install, Start, and Verify Firewalld

Step 1: Install the firewalld package using your distribution's native package manager:

```
# For RHEL / AlmaLinux / Rocky Linux / Fedora:
sudo dnf install firewalld -y

# For Ubuntu / Debian:
sudo apt update && sudo apt install firewalld -y
```

Step 2: Enable firewalld so it automatically starts at boot, then start the systemd service immediately:

```
sudo systemctl start firewalld
sudo systemctl enable firewalld
```

Step 3: Verify that the service is running properly using both systemctl and the firewall-cmd CLI:

```
sudo systemctl status firewalld
sudo firewall-cmd --state
```

Task 2: Allow Incoming HTTP (Port 80) Traffic

Step 1: Add the HTTP service rule to the firewall. Using the --permanent flag ensures the rule persists across service reloads and server reboots:

```
# Allow the predefined HTTP service permanently:
sudo firewall-cmd --permanent --add-service=http

# (Alternative approach using port definition directly):
# sudo firewall-cmd --permanent --add-port=80/tcp

# Reload firewalld to apply changes from the permanent configuration:
sudo firewall-cmd --reload
```

Step 2: Confirm that the HTTP service or port 80 is active in the current zone:

```
# Check active services:
sudo firewall-cmd --list-services

# Inspect full default zone details:
sudo firewall-cmd --list-all
```

Task 3: Temporarily Block Port 8080 and Test Access

Step 1: By default, Firewalld operates with a drop/reject policy for unspecified ports. To test an explicit block during runtime without saving to permanent storage, apply a rich rule dropping TCP port 8080:

```
sudo firewall-cmd --add-rich-rule='rule family="ipv4" port port="8080" protocol="tcp" drop'
```

Step 2: Start a minimal lightweight Python web server on port 8080 in the background:

```
python3 -m http.server 8080 &
```

Step 3: Test network connectivity. The connection request will time out due to the drop rule:

```
# Test locally (timeout after 5 seconds):
curl -m 5 http://127.0.0.1:8080

# Or test from another network client:
curl -m 5 http://<server-ip>:8080
```

Step 4: Terminate the background Python web server once testing is concluded:

```
kill %1
```

Task 4: List Active Zones and Allowed Services/Ports

Step 1: Query all currently bound and active network zones:

```
sudo firewall-cmd --get-active-zones
```

Step 2: Inspect the specific services, ports, and protocols currently permitted under the public zone:

```
# List allowed services in the public zone:
sudo firewall-cmd --zone=public --list-services
```

```
# List allowed ports in the public zone:
sudo firewall-cmd --zone=public --list-ports

# View complete public zone configuration overview:
sudo firewall-cmd --zone=public --list-all
```

Task 5: Remove the Custom HTTP Rule and Verify

Step 1: Remove the previously configured HTTP rule permanently, then reload the firewall daemon to activate the removal:

```
# Remove HTTP service permanently:
sudo firewall-cmd --permanent --remove-service=http

# (If added via port instead):
# sudo firewall-cmd --permanent --remove-port=80/tcp

# Reload to apply changes immediately:
sudo firewall-cmd --reload
```

Step 2: Verify that 'http' (or port 80) no longer appears in the list of active services:

```
# Check active services:
sudo firewall-cmd --list-services

# Confirm in the full zone list:
sudo firewall-cmd --list-all
```